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On the Potential of Liquid-piston Near-Isothermal Compressed Air for Inter-seasonal Electricity Storage

Literature Review

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1. Introduction

The global push towards achieving net-zero targets has significantly heightened the demand for renewable, reliable, and secure electricity sources, resulting in the creation of a 318 GW energy capacity till 2024.¹ In this context, grid-scale energy storage (ES) plays a crucial role, particularly inter-seasonal energy storage, which is essential for ensuring a consistent and stable electricity supply across varying seasons. This is essential for ensuring a consistent and stable electricity supply across varying seasons. Inter-seasonal storage serves as a key component in the broader electricity transition, bridging the gap between electricity generation and the electrification of various industries, thereby supporting a continuous and stable output rate. The main directions of practical grid-scale ES technologies include: (a) pumped-hydro electricity storage (PHES); (b) hydrogen electricity storage (HES); (c) electrochemical electricity storage (EES); and (d) thermal-mechanical electricity Storage (TMES).^{23,4 3,4}

These technologies are explored in detail in **section 2**, providing insights into their capabilities, applications, and potential contributions to a sustainable, low-carbon energy future. Each of these storage systems offers unique advantages and challenges that must be addressed to fully integrate renewable energy into the grid while meeting seasonal demand fluctuations.

1.1 Aims & Objectives

A new potential solution toward liquid piston near-isothermal electricity storage will be discussed to solve previous limitations of past designs. In this literature, main storage types are classified by forms of energy to cover gravity (mechanical), chemical, electro-chemical, and thermo-mechanical energies include different technologies like pumped thermal, liquid air, and compressed air electricity storages. After that, a selection bases on techno-economic indicators to specify more solutions at compressed air electricity storages with respective advantages and limitations.

2. Overview of candidate technologies for inter-seasonal electricity storage

2.1 Pumped hydro electricity storage

Pumped-hydro electricity storage (PHES) is the most mature ES candidate with a total installed capacity of 158 GW in 2023,² which represents more than 90% of global grid-scale energy storage capacity. In 2050, it is expected to grow to 420 GW which means it still be the main technology in grid-scale energy storage area due to its maturity, high discharge efficiency, and long lifetime. In a PHES plat, electricity is consumed to pump water from a low-attitude positions to a higher position to

store energy as potential energy, which is released through a turbine to generate electricity during discharge. Usually, the vessel used to store the water is using the topographies to have reservoirs situated at different elevations. In this case, this technology's development limitation is topography with high water capacity and enough elevation differences, besides that, building time and low energy density of 2 kWh per cubic meter also be the concerns in the future.^{2,5}

2.2 Hydrogen electricity storage

Hydrogen electricity storage (HES) is storing high-pressure hydrogen or other hydrogen-involved compounds to achieve high energy density, and the source of hydrogen is the main concern. Hydrogen production per year is over 50 million tons but with 95% of them are produced with using fossil fuels which lead to sustainable concerns, so using electricity to electrolysis water to form hydrogen can be the solution but also bring economical concerns and lead to low round-trip efficiency about 30% to 40%, in this case, its capacity is less than 100 MW in 2020.^{6,7} However, due to its high energy density (1 kg of hydrogen can store about 33 kW electricity), it still be considered as the main candidate of grid-scale energy storage if electrolyzer's efficiency and power can be developed better.

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2.3 Electro-chemical electricity storage

Electro-chemical electricity storage (ECES) is a good technology candidate in energy storage for both grid-scale storage and commercial storage. There are two main categories include sealed and flow batteries. In sealed battery, electrodes are built by active materials with an ion-conducting electrolyte separator. In flow battery, electrodes are two liquid form electrolytes and been circulated and stored outside of the system. ECES always be considered to have very high round-trip efficiency about 85% to 95% with no limitation of location, in this case, it is used more at commercial applications like EVs (electric vehicles). However, its storage duration time is not long to enough to be considered for inter-seasonal or grid-scale energy storage.^{2,9,10}

2.4 Thermo-mechanical electricity storage

Thermo-mechanical electricity storage (TMES) is based on transformation between electricity and mechanical or thermal energy. During the process, leveraging changes in temperature, pressure, or phase transitions of materials, then convert it back into electricity when needed. At this paper, a few techs of TMES system will be discussed below.^{2,11,12}

2.4.1 Pumped Thermal electricity storage

Pumped Thermal electricity storage (PTES) is a large-scale energy storage technology that converts electrical energy into thermal energy during charging and turn it back into electricity as Carnot Batteries. The thermal stores consist of a hot and cold a reservoir, typically at 500 to 1000°C and -150 to 0°C, respectively (with materials like molten salts and ceramics).^{13,14} Then, the challenge is the efficiency of thermal storage to stop leaking heat via the surface of vessel and will increase cost of the vessels and will increase cost of storage. Moreover, the discharging is using the fluid from hot reservoirs to expand working gas like CO₂ through a turbine to generate the electricity back and the fluid is sent back to cold reservoirs, and this process is called Joule-Brayton cycle. However, this process is not very efficient, with a round-trip efficiency to be 50-60%.¹⁵

2.4.2 Liquid Air electricity storage

Liquid Air electricity storage (LAES) is the way to using electricity to transfer air which typically be nitrogen from gas phase to liquid phase and be stored in vacuum pressure to stay them in liquid more effectively, then they will release and heat up to create high pressure stream to generate electricity back like **Figure 1** illustration. It is still a developing tech which only installed about 5 MW.¹⁶ Due to its operating conditions are rigorous, its capital and operating costs which to build the storage vessel and maintaining energy are so high that cannot compete with others, meanwhile, this long-term technology will be challenged due to its vacuum storage pressure.¹⁷

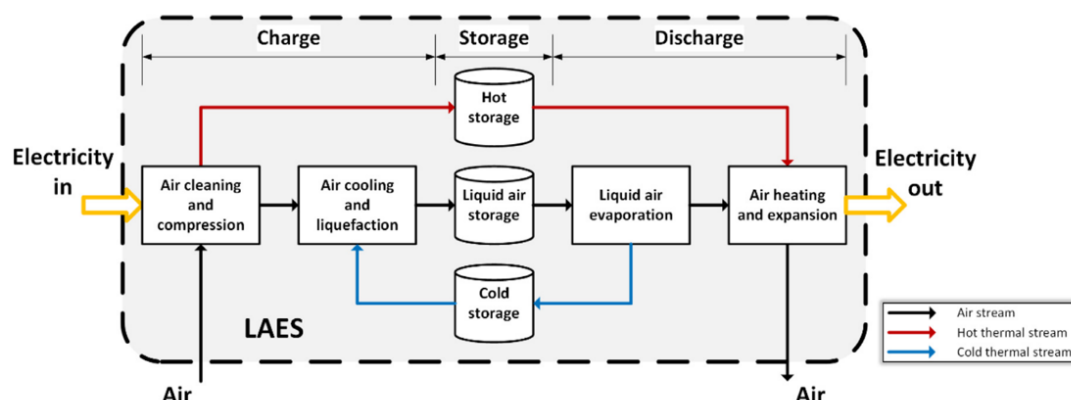


Figure 1: Flow Diagram of Liquid Air Energy Storage. Adapted from Andrea Vecchi, A review on technology state-of-the-art, integration pathways and future perspectives

2.4.3 Compressed Air electricity storage

Compressed Air electricity storage (CAES) is the way to compress air without phase changes. However, like LAES, there is a heating and cooling system which differentiate 3 types of CAES: (a) conventional CAES; (b) adiabatic CAES; and (c) isothermal CAES. Due to its storage vessel usually be natural underground cavern and low storage pressure, its duration is high to across seasons and good cycle efficiency about 70% for adiabatic CAES which made it to be a very competitive candidate for inter-seasonal storage.

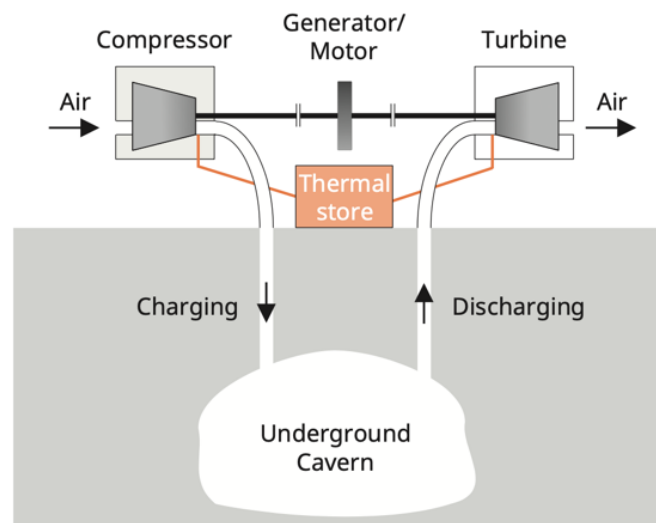


Figure 2: Process of Compressed Air Energy Storage. Adapted from Oliver Schmidt, *Monetizing Energy Storage*

However, in adiabatic CAES, during the compression, the cooling and heating system is the thermal store to store generated heat from compression and warm up pressure-decreased expansion, and the thermal storage just like PTES which means the decay of heat is unavoidable. In this case, the part of heat decayed will be covered by using natural gas burning which will increase costs and release greenhouse gases (GHG).^{2,17,18}

2.5 Techno-economic indicators

To select inter-seasonal energy storage, the discharging duration, energy density, electricity capacity, and power-specific investment cost of each technology will be the main indicators to be considered. By following the definition of inter-seasonal energy storage, the candidate's duration is higher than 2 months with GWh capacity. **Figure 3** shows a simple distribution of main 5 energy storage categories, where only part of CAES, PHES, and HES can be considered as inter-seasonal energy storage technologies.

For PHES, it is the most developed energy storage technology with good discharging duration and

capacity but its energy density highly related with the height for the matter to drop, in this case, to increase its energy density, it will require rigorous geological conditions that increase high capital cost of building. Conversely, HES has high density 600 kWh per cubic meter for pure hydrogen via electrolyzing water without further reactions, and the storage of hydrogen can be long enough, however, its power-specific investment cost is about 5000 USD/kW what is greater than other two candidates.¹⁹

Then, CAES will be the left candidate which shows its unique characteristics like less-rigorous geological requirements, low power-specific investment cost 1000 USD/kW, medium energy density 8 kWh per cubic meter.^{20,21}

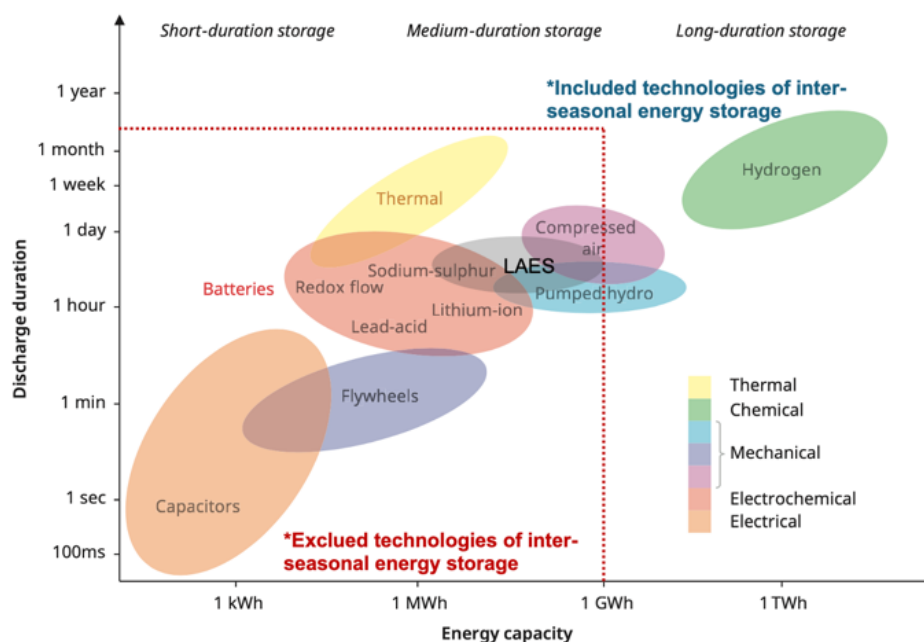


Figure 3: Discharge duration and electricity capacity of various technologies of main five energy storage categories. Adapted from Sterner, *Energiespeicher—Bedarf, Technologien, Integration*

3. Limitations of Conventional & Adiabatic CAES

3.1 Conventional CAES

Conventional CAES or diadiabatic CAES (D-CAES) does not have a thermal storage system to save energy instead of using separated coolers and fossil fuels during compression and expansion process. The very first D-CAES plant is using the Huntorf Plant and located at Germany to store air at 44 to 70 bars in the vessel which volume is $3.1 \times 10^5 m^3$ with round-trip efficiency of 42%.¹⁷ During the developments, a plant is trying to use part of low temperature flow at the expansion stage to cool down the compression flows and called the McIntosh plant, in this case, the McIntosh plant increased round-trip to 54%.¹⁷ The potentials for D-CAES to be economically profitable and environmentally

friendly can be achieved when fuel prices higher than \$7.6 per GJ and pipeline lengths lower than 50 km.^{22,23} However, due to the nature of this system, there is high amount of heat what cannot be used, the needed heat is provided by fossil fuels what will create high emissions, and its requirements toward geological conditions are higher other systems.²⁴

3.2 Adiabatic CAES

Adiabatic CAES (A-CAES) is different with D-CAES by introducing a thermal storage system to increase the thermal efficiency which can be generally illustrated by **Figure 4**. It is diversified to two directions: (a) Advanced Adiabatic CAES (AA-CAES) and (b) Low-temperature Adiabatic CAES (LTA-CAES) bases on choosing to use one single-stage to compress air with 520°C or choosing to use multiple-stages to compress with multiple heat exchangers to main flows at low temperatures.²⁵

Figure 4 is demonstrating a two-stage compression and a two-stage expansion design.²⁶

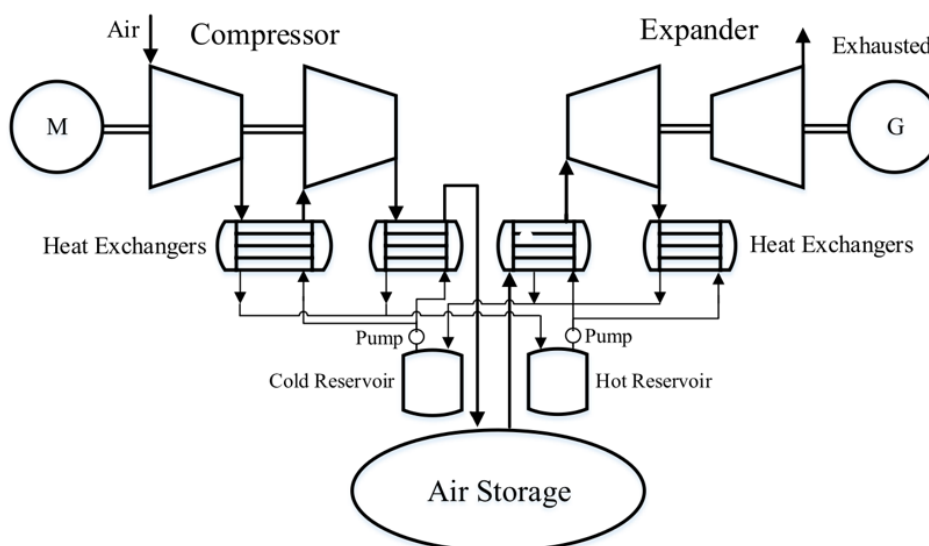


Figure 4: A Type of A-CAES (LTA-CAES) Process Illustration. Adapted from Andreas V Olympios, *Progress and prospects of thermo-mechanical energy storage—a critical review*

In the plant design of AA-CAES, its round-trip efficiency is defined simulatively 70% with 95% thermal energy storage efficiency.²⁷ However, there are difficulties about designs of high-temperature flow in compression system and thermal energy storage system due to nature thermal emission. In this case, LTA-CAES introduce multiple heat exchangers to ensure the low temperature is maintained to decrease the design's complexity and the emission of thermal energy storage with 68% round-trip efficiency.²⁸ The relationship between the number of heat exchangers and the round-trip efficiency has been researched that higher heat exchanger effectiveness will have higher the round-trip

efficiency under no-pressure-loss assumption.²⁹ However, LTA-CAES's thermal storage system's complexity is high that requires high capital cost of heat exchanger units and pipe, meanwhile, its efficiency 68% can be improved better.²⁸ However, there is still additional heat source like fossil fuel which is required to cover the expansion heat requirement.

3.3 Isothermal CAES

Isothermal CAES (I-CAES) is the technology that trying to achieve isothermal compression and expansion by continuous effective heat transfer during charging and discharging. Its property is about continuous effective heat transfer which doing heat transfer during compression and expansion immediately and it distinguishes with A-CAES which happens periodically. That leads I-CAES round-trip efficiency theoretically higher than A-CAES's which is 70% to 80%.³⁰

There are two types of I-CAES: (a) Closed I-CAES and (b) Open I-CAES. For the closed I-CAES, **Figure 5** shows a spray water as the injection of liquid during compression of air to have continuous effective heat transfer, then the air-water mixture will be separated to stored high-pressure air and water individually. For the open I-CAES, it consists with 2 alternating cylinders that can make the heat transfer more continuous, but it leads to be open and not air storage will make its duration to be shorter. Additionally, ocean based I-CAES is a new direction to explore by using ocean water to do heat transfer as media, but it is still a concept.^{31,32}

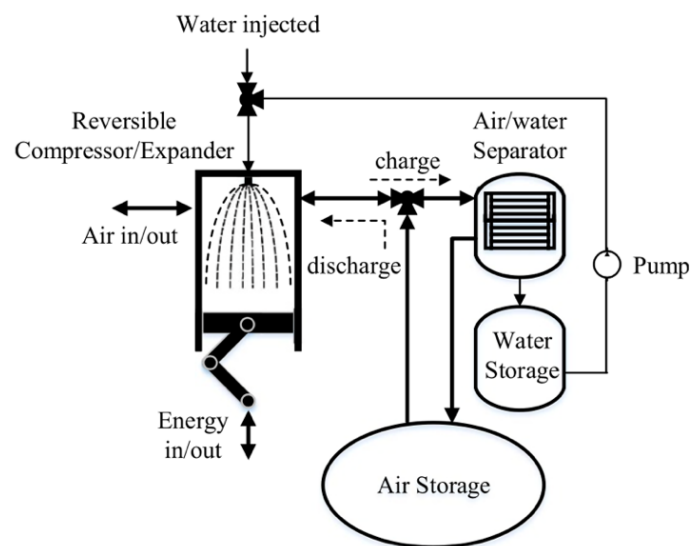


Figure 5: Closed I-CAES Process to Achieve Isothermal Compression and Expansion by injecting water into one cylinder during piston compression. Adapted from Andreas V Olympios, *Progress and prospects of thermo-mechanical energy storage—a critical review*

The limitation of I-CAES is the heat transfer during compression and expansion, however, there is

opportunity to overcome. The industrial solutions would be using more advanced heat exchanger, water sprays, and liquid piston technology. The one that **Figure 5** illustrating is using water sprays during the compression and expansion, however, the design of it leads its cost to be high under high airtight requirements and system complexity, meanwhile, the heat of air can be transferred effectively but the heat of pump include piston cannot be transferred easily. In this case, the part of this heat loss makes its round-trip efficiency can be improved better. A new is using liquid as piston of pump inside heat exchangers can solve this complicated problem and will be discussed with less heat loss, longer duration, less greenhouse gas emissions, and high round-trip efficiency.^{33–35}

4. Technological solutions for near-isothermal CAES

4.1 Introduction to liquid piston near-isothermal CAES

Due to the almost heat generation is from compression, the design of compressor needs to be modified to achieve near-isothermal process by increasing its heat transfer efficiency. Liquid piston (LP) based CAES is using liquid as piston instead of solid one to push the gas up from the bottom through the cylinder, and this solution not only solve the considerations of heat transferring but also the gas leakage. Meanwhile, it is necessary to discuss about operating principles and thermodynamic background.

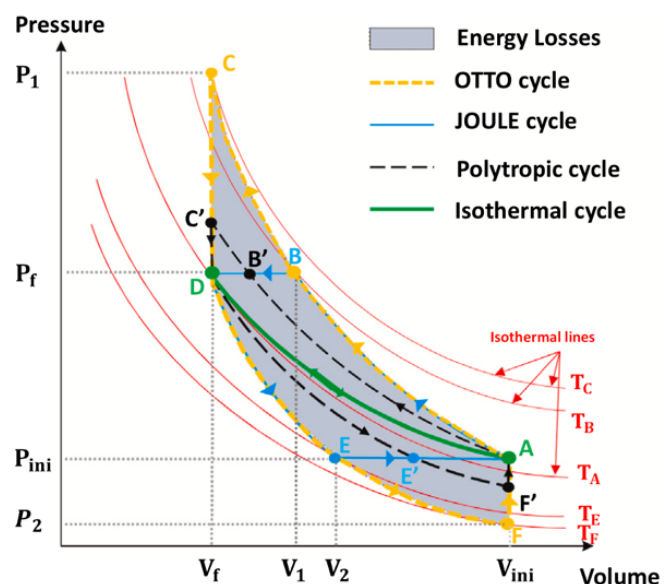


Figure 6: Pressure-Volume diagrams of Air compression and expansion of different thermodynamic cycles include OTTO cycle, JOULE cycle, Polytropic cycle, and Isothermal cycle. Adapted from S. Lemoufuet, *Investigation and optimisation of hybrid electricity storage systems based on compressed air and supercapacitors*

For operating principles, during this LP process, there are two stages include charging stage and

discharging stage. For charging stage, the electricity is used to power up hydraulic pumps to fill the liquid into the cylinder where is full of gas initially, during the time, the gas is compressed from its start point P_0 to the setting point P_c , and the ratio between P_c and P_0 is called the compression ratio (CR). Another stage is releasing the gas from the gas storage vessel, due to the nature of gas, it will expand to push the piston or the liquid back through a turbine to generate electricity during discharging stage.

Thermodynamic inside this technology requires to be specified. To begin with, there are air compression and expansion of thermodynamic cycles what also be presented in **Figure 6**, by checking the area of energy losses, isothermal cycle takes the best performance and following with JOULE cycle, and OTTO cycle.^{36–40} The reason of ranking is the assumption of isothermal process, and it is way too ideal and can only be achieved to be near isothermal. The formulas of work during compression and expansion inside LP I-CAES are Eq (1) and (2):

$$W_c = \int_{V_0}^{V_c} \underbrace{(P(t) - P_0)dV}_{\text{compression flow}} + \underbrace{\left(1 - \frac{1}{CR}\right)(P_c V_c - P_0 V_0)}_{\text{isobaric cooling}} + \underbrace{(P_c - P_0)V_{iso}}_{\text{pumping}} \quad (1)$$

$$W_e = \int_{V_0}^{V_e} \underbrace{(P(t) - P_f)dV}_{\text{expansion flow}} + \underbrace{(P_c - P_f)V_0}_{\text{pumping}} \quad (2)$$

After the storage, during using the compressed air, the potential energy of it can be expressed Eq (3):

$$E_p = \underbrace{(P_c - P_0)V_{iso}}_{\text{pumping}} + \underbrace{P_0 V_0 \left(\ln(CR) + \frac{1}{CR} - 1 \right)}_{\text{Isothermal expansion}} \quad (3)$$

Its compression efficiency η_c and expansion efficiency η_e are defined respectively as

$$\eta_c = \frac{E_p}{W_c} \quad (4) \quad \& \quad \eta_e = \frac{W_e}{E_p} \quad (5)$$

There is a sensitive test about the effect of the compression ratio (CR), the cylinder geometry, and the piston velocity or liquid velocity on η_c . There is enough evidence to show a higher CR (from 5 to 50) will have a lower η_c (from 87.9% to 68.9%), in this case, a low CR will be preferred with the limitation of 5 to have higher energy density. By looking back the ratio $P_c:P_0$, if P_0 is set to be 2 or 3 atm, P_c will be 10 or 15 atm what would lead to high pressure situation. Piston velocity analyzation brings a finding that it has small impact on η_c about 5% under high pressures after achieving 0.05 meter per second, in this case, a longer cylinder with smaller diameter or high L/D would be preferred to have high efficiency, but smaller diameter will bring the heat transfer concerns out since the most

of heat is transferred by convections on the interface of liquid and gas. Meanwhile, it is the main challenge of LP I-CAES technology to do tradeoff between round-trip efficiency and energy density, at this paper, there are two main solutions will be present below.^{41–44}

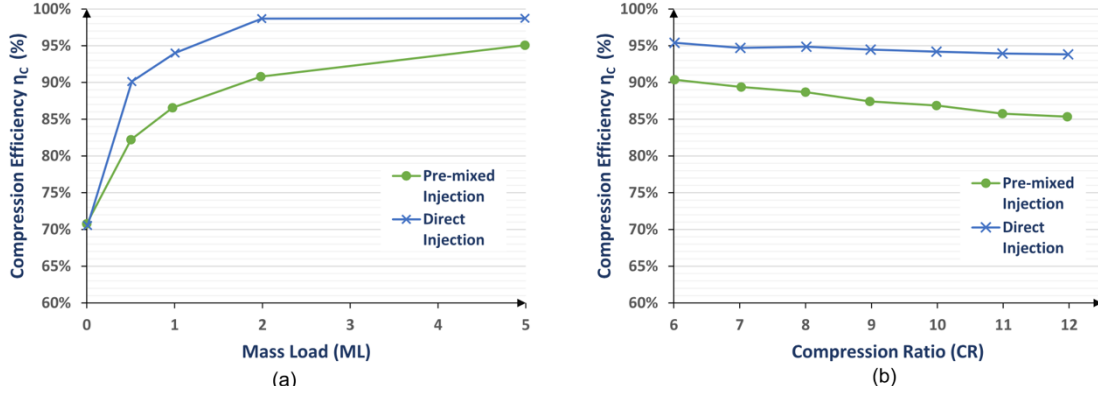


Figure 7: (a) compression efficiency-ML under CR is set to be 10 & compression efficiency-CR diagrams under MDD is set to be 20 micrometer diagrams of water-spray LP model. Liquid piston compression efficiency with droplet heat transfer

4.2 A method to combine water-spray and liquid piston

Water spray liquid piston I-CAES is a combination technology of the **section 3.3** water-spray I-CAES and LP I-CAES that the water is injected at the top of column with the LP at the bottom. During the process, operation parameters are defined to be the mean droplet diameter (MDD), mass load between water sprayed and dry air inside the column (ML), and injected water temperature. **Figure 7** shows the relationship of ML to efficiency and CR to efficiency respectively by a 1-D simulation, which proves that higher ML will have better heat transfer efficiency for both two main types from 70% to 98% range, but CR's effects on efficiency are not strong as ML but relatively low CR (maximum CR will be set as 10) can bring better efficiency but also low energy density, in that case, it will face some industrial challenges.³⁰

4.3 A method to combine heat exchanger and liquid piston

There is another way which is using structure of the liquid piston vessel. In chemical and engineering-based industries, heat exchanger is a well-used unit to do heat transfer, in this progress, a design is using the structure of heat exchanger as the vessel of compressor. In the **Figure 8**, a plate heat exchanger, liquid in the water reservoirs will be pumped to first plate will is full of air and increased the air pressure and decrease the space of air, then the air in first plate is compressed to the third plate, fifth plate, and seventh plate, meanwhile, another flow of water will be pumped to the cooling system at the second, fourth, sixth, and eighth plate. During the process, the heat transfer happens at the convection of liquid-air interface, the walls between compression plates and cooling plates, and

the wall to ambient air.

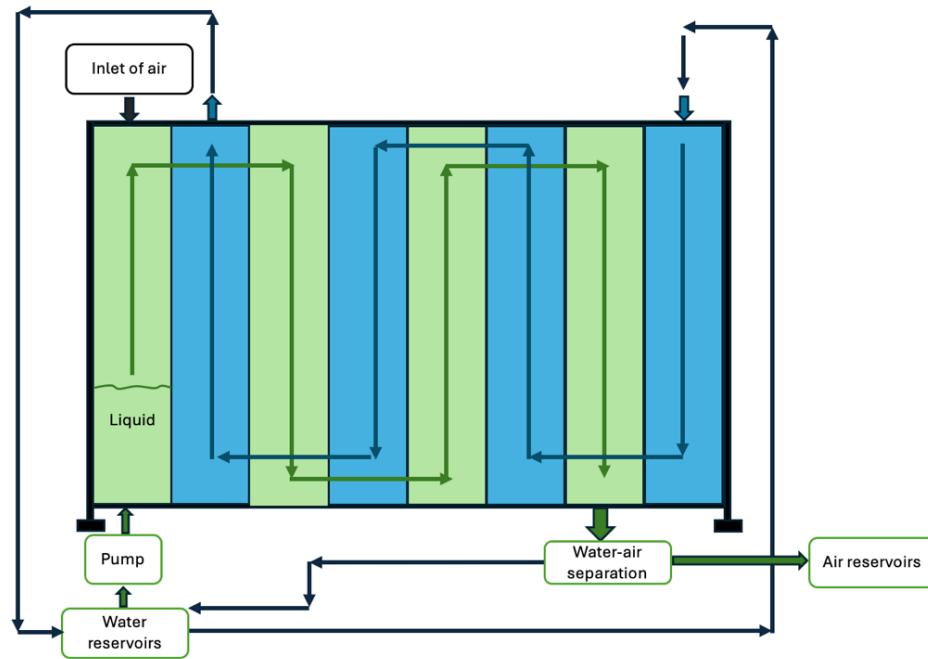


Figure 8: Plate structure heat exchanger as liquid piston compressor.

However, this design is a developing technology with rare simulations and experiments to find its operational parameters like L/D ratio, compression ratio, and fluid patterns and to evaluate its compression efficiency. In this case, it is necessary to develop a computational model to simulate it to have its fluid behaviors, heat transfer efficiency, and compression efficiency.

5. Computational fluid dynamics

Due to the complexity of the above technology, especially for flow patterns, computational fluid dynamics (CFD) can be used to analyze potential fluid behaviors, heat transfer, pressure field inside the structure before an actual experiment is settled to examine its actual performance in laboratory scale.

CFD is using numerical models to analyze fluid flows problems. The steps of solving problem are: (a) mathematical modeling include governing equations and turbulence equations; and (b) geometry, meshing, and boundary conditions. In this case, the beginning of using CFD is designing the aim that get what extent level of data which defines how accurate of results by having decisions on dimension of modeling, assumption of fluid properties, geometry of model, meshing of simulations, CR number, the approach of volume of fluid, and so on. Then, mathematical modeling can be established, at this review, it will not be discussed, but the additional equations will relate with heat transfer behaviors.⁴⁵

6. Conclusion

In conclusion, this paper discussed few main storage types by energy categories, types of thermomechanical direction and types of isothermal compressed air electricity storage to decide to focus on isothermal compressed air electricity storage because of its investment cost, potential round-trip efficiency development, duration time, and energy density. A critical discussion is made between diadiabatic, adiabatic, and near-isothermal compressed air electricity storage that finding high economical and technical potentials of near-isothermal. Following it, a liquid-piston design is the main direction, however, with the tradeoff between energy density and efficiency, the solution is focusing on solving the limitation which is the ratio of compressor length and diameter can be solved by two main designs include the liquid-piston with water spray method and the liquid-piston with heat exchanger method. Ideally, the liquid-piston with water spray method can achieve 98% compression efficiency. For the liquid-piston with heat exchanger method, a possible evaluation consists of a computational analysis by computational fluid dynamics and an experiment to find out its fluid behaviors, heat transfer efficiency, and compression efficiency.

7. Research Plan

Figure 9 has a structured research plant to help you understand. After finishing this literature review, health and safety test will be fined in March with the start of pre-simulation learning, in this case, CFD learning will be done to help understanding toward the tool. Meanwhile, an simulation plan will be created to guide. After that, it is exam period and consists with exam preparation during April and exam during May. A proper start of research is in June to have simulation inductions and simulation which last from June to August. Report writing will be focused in early July with continuous modifications with co-supervisor. After all simulations and writing are done, the preparation of presentation will start in middle of August for poster and in last week of September for the presentation.

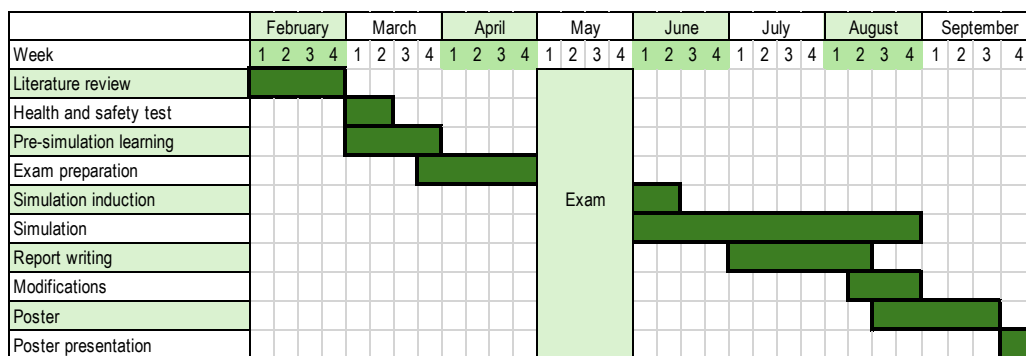


Figure 9: Gantt charts of research plan

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